CHAPTER 19: TIME-VARYING TREATMENTS & STRATEGIES

Causal Inference in Longitudinal Telco Customer Churn & Retention Programs

1. Longitudinal Treatment Framework in Telco Retention

In industrial telecommunications analytics, customer retention is rarely a single point intervention. Subscribers receive time-varying exposures A_k over multiple monthly billing cycles k = 0, 1, ..., K (e.g., $A_k = 1$ if a subscriber receives a targeted loyalty discount, bandwidth upgrade, or proactive service credit in month k; $A_k = 0$ otherwise). Concurrently, customer behavior and network experience evolve as time-varying covariates L_k (e.g., monthly data consumption, customer service ticket volume, network quality alerts, and billing complaints). The ultimate business outcome Y represents 12-month account churn status (Y = 1 if churned, Y = 0 if retained).

Notation / Variable	Causal Role	Telco Operational Definition
A_k	Time-Varying Treatment	Retention offer or billing discount applied in month k
$\bar{A}_k = (A_0, ..., A_k)$	Treatment History	Sequence of monthly retention interventions up to month k
L_k	Time-Varying Covariates	Support tickets, data usage, dropped calls, bill shock in month k
$\bar{L}_k = (L_0, ..., L_k)$	Covariate History	Accumulated subscriber usage and service history up to month k
Y	End-of-Horizon Outcome	Final account churn status at end of evaluation period (K + 1)

2. Static vs. Dynamic Treatment Strategies (g)

Evaluating time-varying treatments requires explicitly defining multi-period **treatment strategies** (also called protocols or regimes). A treatment strategy $g = (g_0, g_1, ..., g_K)$ specifies the treatment assignment rule $g_k(\bar{A}_{k-1}, \bar{L}_k)$ at each month k:

> **Key Strategy Taxonomy in Customer Retention:**
> • **Static Always-Treat ($\bar{a} = 1$):** Apply retargeting discount every month regardless of usage ($\bar{a} = (1, 1, ..., 1)$).
> • **Static Never-Treat ($\bar{a} = 0$):** Never offer proactive retention discounts ($\bar{a} = (0, 0, ..., 0)$).
> • **Dynamic Rule-Based Strategy (g^{dyn}):** Trigger a 15% bill discount in month k IF support calls $L_k \geq 2$ or monthly usage drops by >20%, otherwise no discount ($A_k = 0$).

The primary causal estimand is the Average Treatment Effect (ATE) contrasting counterfactual churn risks under two sustained strategies g and g': **$ATE = E[Y^g] - E[Y^{g'}]$**. Unlike single-point interventions, comparing time-varying strategies requires accounting for how treatments alter subscriber covariate trajectories over time.

3. Sequentially Randomized Experiments & Observational Data

In a **sequentially randomized experiment**, treatment A_k at each month k is randomly assigned with a known probability that depends on prior treatment and covariate history $(\bar{A}_{k-1}, \bar{L}_k)$. In observational telco databases, marketing teams naturally target retention offers to high-risk or complaining subscribers. Observational logs mirror sequentially randomized trials, except treatment assignment probabilities $P(A_k = 1 \mid \bar{A}_{k-1}, \bar{L}_k)$ are unknown and must be estimated from historical data.

4. Core Identifiability Conditions for Time-Varying Treatments

To identify counterfactual outcomes $E[Y^g]$ from observational longitudinal data, three sequential identifiability conditions must hold at every time step k = 0, 1, ..., K:

Identifiability Condition	Mathematical Statement	Telco Domain Requirement & Verification
1. Sequential Exchangeability	$Y^g \perp A_k \mid \bar{A}_{k-1} = \bar{a}_{k-1}, \bar{L}_k = \bar{l}_k$	No unmeasured confounders U_k influencing both month k offer assignment A_k and final churn Y. All retention targeting factors captured.
2. Sequential Positivity	$P(A_k = a_k \mid \bar{A}_{k-1}, \bar{L}_k) > 0$	For all subscriber profile histories, there is a non-zero probability of both receiving and not receiving the retention offer at month k.
3. Sequential Consistency	$Y^g = Y$ if $A_k = g_k(\bar{A}_{k-1}, \bar{L}_k)$ $\forall k$	Retention interventions are well-defined (e.g., standard 15% discount for 30 days) with no multiple ill-defined versions.

5. Structural Causal DAGs & SWIG Representation

Time-varying causal structures can be represented using Directed Acyclic Graphs (DAGs) and Single World Intervention Graphs (SWIGs). Figure 19.1 below depicts a two-period longitudinal retention system with treatment A_k, time-varying confounders L_k, unmeasured factors U_k, and final churn Y.

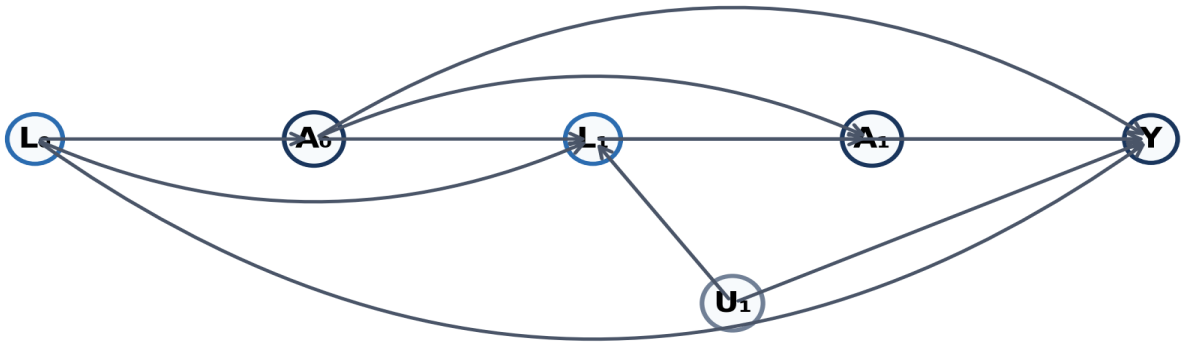


Figure 19.1: Longitudinal Causal DAG showing time-varying confounding and treatment history ($A_0 \to L_1 \to A_1 \to Y$).

> **SWIG Identifiability Insight (Chapter 19.5):**
> Splitting treatment nodes A_k into observational A_k and intervention value a_k reveals that static strategies $\bar{a}$ are identified whenever $Y^{\bar{a}} \perp A_k \mid \bar{A}_{k-1}, \bar{L}_k$. However, dynamic strategies $g(\bar{L}_k)$ require that unmeasured baseline factors W_0 do not simultaneously affect A_0 and future covariates L_1.

6. Strategy Comparison & Methodological Takeaways

Dimension / Property	Static Strategy ($\bar{a}$)	Dynamic Strategy ($g(\bar{L}_k)$)
Rule Definition	Fixed sequence independent of L_k (e.g., Always Treat)	Adapts to subscriber status L_k (e.g., Treat if calls ≥ 2)
Identifiability Check	Generalized Backdoor / SWIG $Y^{\bar{a}} \perp A_k \mid \bar{A}_{k-1}, \bar{L}_k$	Requires $Y^g \perp A_k \mid \bar{A}_{k-1}, \bar{L}_k$ (stricter graph rules)
Telco Business Value	Evaluates global campaign policies across all users	Evaluates personalized, trigger-based retention rules
Estimation Methods	IPW, Longitudinal G-Formula, G-Estimation	Longitudinal G-Formula, IPW for Dynamic Regimes